

FBR Digital Invoicing — Frontend Requirements Report

For	UI/UX Designer
Project	FBR Z — Pakistan Federal Board of Revenue E-Invoicing Portal
Date	May 2026

1. Project Overview

This is a web application used by Pakistani businesses to submit tax invoices to FBR (Federal Board of Revenue) under S.R.O. 69(I)/2025. The system handles invoice creation, submission to the FBR gateway, offline queueing, product management, customer management, and sandbox certification testing.

Current Tech Stack:

- React 19 frontend (single-page application)
- REST API backend (Node.js + TypeScript)
- Authentication: JWT stored in localStorage
- Google Translate integration (English, Urdu, Spanish, French, Arabic)

Design Language (Current):

- Bootstrap 5 base
- Custom card shadows (bg-dashboard-shadow, bg-dashboard-shadow2)
- Custom button classes (buttonsubmitinvoice, buttonsave, buttonpreview)
- Icons: React Icons library (Feather, Material Design, Ionicons)

2. Application Structure

Route Map

```
/ ————— Login
/forgot-password — Forgot Password Step 1 (email entry)
  /reset-password — Step 2 (OTP/code)
  /new-password — Step 3 (new password)

/dashboard ————— Dashboard (home after login)
/invoice ————— Invoice List
  /invoice/add — Add New Invoice
  /invoice/upload — Upload Invoice (JSON)
  /invoice/upload/preview — Preview before submission
  /invoice/offline-queue — Offline Queue
/company-profile — Company Profile
/customers ————— Customer Address Book
/products ————— Product / HS Code Mapping
/staff ————— Staff Directory
```

```
/services — Services page (informational/marketing)
/sandbox — FBR Sandbox Testing
/settings — FBR API Settings
/support — Support / Help
/account/* — Account sub-pages:
  /account/edit-profile
  /account/change-password
  /account/two-factor
  /account/language
  /account/privacy
  /account/terms
  /account/deactivate
```

Global Layout

Authenticated pages use a persistent left sidebar + main content area.

Sidebar contains:

- Logo (full or icon-only when collapsed)
- Nav items: Dashboard, Invoice (expandable), Company Profile, Customers, Services, Products, Staff, Sandbox
- Bottom section: Settings, Support, Logout
- "Powered by Techionik" branding
- Collapse/expand toggle button (desktop)
- Mobile hamburger toggle

Sidebar states:

- Expanded (default, ~270px wide)
- Collapsed (icon-only, ~80px wide)
- Mobile drawer (slides in over content, close button)

Offline Banner: A persistent top banner (appears across all pages) showing:

- "You are offline" warning with pending queue count
- "Reconnected — processing queue" when connectivity returns
- Goes away when online and queue is empty

3. Screen-by-Screen Requirements

3.1 Login Screen

Purpose: Authenticate user into the portal.

Layout: Split two-column

- Left: login form
- Right: decorative illustration image

Elements:

- Application logo (top-center of left panel)
- Subtitle: "FBR Digital Invoicing"

- Heading: "Login"
- Email input (with user icon prefix)
- Password input (with lock icon prefix, show/hide toggle button)
- "Forget Password?" link (right-aligned)
- Primary button: "Login Now" (full-width, shows spinner while loading)
- Secondary button: "Continue Without Login" (dev mode only — remove in production)
- Social login placeholders: Google, Apple, Microsoft icons (currently non-functional links — designer should note these need real integration or removal)
- "Don't have an account? Sign up" (currently non-functional — note for designer)
- Error message (inline, red)
- Success message (green alert, shown after successful password reset redirect)
- Footer: "Powered by Techionik" with icon

States to design for:

- Default
- Loading (button spinner, inputs disabled)
- Error (inline error below password)
- Success (green alert)

UX notes:

- On successful login → redirect to /dashboard
- If already logged in → redirect to /dashboard immediately
- Back button is blocked on all authenticated pages (users must use sidebar nav)

3.2 Forgot Password Flow (3 Steps)

Step 1 — Email Entry

- Single email input
- "Send Code" button
- Link back to login

Step 2 — Code Verification

- OTP/code input
- "Verify" button
- Resend code option

Step 3 — New Password

- New password input
- Confirm password input
- "Reset Password" button
- On success: redirect to login with success message

UX notes: These are public pages (redirect to dashboard if already logged in).

3.3 Dashboard

Purpose: Overview/home page. Shows key stats, recent invoices, top products, top customers, and a sales chart.

Top Stats Row (3 cards):

Card	Value	Sub-text
Invoices	Count of all invoices	"No of Invoices" + total PKR amount
Customers	Unique customer count	"No of Customers"
Products	Total product count	"No of Products"

Sales Chart:

- Bar chart (Recharts BarChart)
- Period selector: Daily / Monthly / Quarterly / Yearly (toggle buttons or tabs)
- X-axis: period labels, Y-axis: invoice amounts
- Loading state while fetching

Top Invoices Table:

- Paginated (6 per page) with Prev/Next arrows
- Sorted by amount (highest first)
- Columns: FBR Invoice No, Type, Buyer, Date, Amount, Status badge
- Status badges: SUBMITTED (green), UPLOAD_FAILED (red), DRAFT (gray)

Top Products Table (left half):

- 5 rows max
- Columns: Product Name, HS Code, Tax Rate
- Search input

Top Customers Table (right half):

- 5 rows max
- Columns: Customer Name, NTN/CNIC, Invoices (count)
- Derived from invoice buyer data (not the customers address book)

Loading states: Individual spinners per section.

UX notes:

- Dashboard is the first page users see after login
- Data loads on mount; chart reloads when period changes
- Empty states needed for each section

3.4 Invoice List (/invoice)

Purpose: View, search, sort, download, and delete invoice records.

Header Stats Row (2 cards):

- Total invoices count
- Total customers count

Table Controls:

- "Upload Invoice" button → /invoice/upload
- "Add Invoice" button → /invoice/add
- Select All toggle (custom radio-style button with checkmark)
- Delete button (bulk delete selected)
- Download button (bulk download PDFs; single = direct download, multiple = ZIP)
- Search input (free-text: matches invoice ref, buyer name, date, FBR number)

- Sort dropdown: Newest / Oldest / Buyer A→Z / Buyer Z→A / Invoice No A→Z / Invoice No Z→A
- Rows per page: 10 / 20 / 50 / 100 / All

Table Columns:

- Invoice No (with checkbox, shows invoice_ref_no or INV-`{last6}`)
- FBR Invoice No (monospace font)
- Buyer Name
- Invoice Date (with calendar icon, DD/MM/YYYY format)
- Amount PKR (formatted with commas, 2 decimal places)
- Status badge (SUBMITTED=green, FAILED=red, DRAFT=gray)
- Actions: Eye icon (view detail modal)

View Detail Modal:

- Slide-in modal (large)
- Fields: Invoice Ref No, FBR Invoice Number, Invoice Type, Invoice Date, Buyer Name, Buyer NTN/CNIC, Amount (PKR), Status, Source, Submitted At
- Footer: "Download PDF" button, "Close" button

Delete Confirmation Modal:

- Counts selected items
- Warning: "This removes them from your local records only. FBR submission records are not affected."
- Cancel / Delete buttons

Pagination:

- Prev / Page X of N / Next
- Only shown when more than 1 page

UX notes:

- Download of multiple invoices produces a ZIP file
- While downloading, button shows spinner and is disabled
- Toast notifications for success/failure of all operations
- Back button blocked (no accidental navigation away from invoice list)

3.5 Add Invoice (/invoice/add)

Purpose: The most complex screen. Create and submit a new invoice to FBR.

Form Structure:

Section 1 — Invoice Header:

Field	Type	Notes
Invoice Type	Select	Sale Invoice, Sale Invoice (Exempt/Zero-Rated), Credit Note, Debit Note, Purchase Invoice (B2B) — loaded from FBR reference API
Invoice Date	Date picker	Required
Invoice Ref No	Text	Optional reference
Sandbox Scenario ID	Select	Only shown in sandbox environment; loaded from FBR sandbox API

Section 2 — Seller Info (auto-filled from Company Profile):

FBR Digital Invoicing — Frontend Requirements Report

Field	Type	Notes
Seller NTN/CNIC	Text	Pre-filled from company profile
Seller Business Name	Text	Pre-filled from company profile
Seller Province	Select	Pre-filled from company profile; FBR province list
Seller Address	Text	Pre-filled from company profile

Section 3 — Buyer Info:

Field	Type	Notes
Buyer NTN/CNIC	Text	Optional
Buyer Business Name	Text	Required
Buyer Province	Select	FBR province list
Buyer Address	Text	Required
Buyer Registration Type	Select	Registered / Unregistered
Quick-fill from Customer	Button/Dropdown	Loads customer from address book to auto-fill buyer fields

Section 4 — Line Items (repeatable, minimum 1):

Each line item has these fields:

Field	Type	Notes
Product	Select/search	Load from saved products (auto-fills other fields)
HS Code	Select	FBR item descriptions list (6-digit codes)
Product Description	Text	Auto-filled from HS code
Sale Type	Select	~28 options from FBR transaction types
UOM (Unit of Measure)	Select	FBR UOM list
Quantity	Number	Required
Rate	Text	Tax rate
Total Values	Number	Calculated
Value of Sales Excl. ST	Number	Required
Fixed/Notified Value	Number	Required
Sales Tax Applicable	Number	Calculated
Sales Tax Withheld at Source	Number	
Extra Tax	Number	Optional
Further Tax	Number	Optional
SRO Schedule No	Select	Optional; from FBR SRO reference
FED Payable	Number	Optional
Discount	Number	Optional

Add/Remove items: "Add Item" button adds a new row; each row has a remove button.

Form Actions:

- **Submit to FBR** — sends to FBR gateway (requires active token)
- **Save Offline** — queues for later upload (shown when offline or user choice)
- **Preview PDF** — generates and downloads invoice PDF without submitting
- **Cancel** — navigates back

Error Display (`FbrErrorDisplay` component):

- Shows FBR API validation errors in a structured list
- Distinguishes FBR field errors from general API errors
- Errors appear inline below the form

Status indicators:

- Online/offline detection — affects whether "Submit" or "Save Offline" is available
- Mock mode indicator (in sandbox: fake invoice numbers generated)

Reference data loading:

- All dropdowns load from FBR bootstrap API on mount
- Falls back to hardcoded options if API unavailable

UX notes:

- This is the most critical screen — invoice compliance with FBR Rule 150R requires all 26 fields
- Seller fields auto-fill on mount from company profile (reduce data entry)
- Selecting a product auto-fills HS code, description, sale type, UOM, rate
- Selecting a customer auto-fills buyer fields
- HS code selection auto-fills description
- PDF preview generated client-side using jsPDF (includes FBR DI logo, QR code, all fields)
- Validation errors from FBR are shown clearly per field

3.6 Upload Invoice (/invoice/upload)

Purpose: Upload an invoice from a JSON file instead of using the form.

Elements:

- File drop zone / file picker (accepts JSON)
- Instructions on expected JSON format
- "Preview" button → navigates to /invoice/upload/preview with parsed data

3.7 Invoice Preview (/invoice/upload/preview)

Purpose: Review a parsed invoice from JSON file before submitting to FBR.

Elements:

- Read-only display of all invoice fields (same structure as Add Invoice)
- "Submit to FBR" button
- "Edit" / "Back" button
- Error display for FBR validation errors

3.8 Offline Queue (/invoice/offline-queue)

Purpose: View and manage invoices that were saved offline and need to be uploaded to FBR. Critical for compliance (Rule 150XC — 24-hour upload deadline).

Top Stats Row (5 cards):

Stat	Color
Pending Upload	Gray
Upload Failed	Red
Submitted	Green
20h Warnings	Yellow
24h Deadlines	Red

Alert Banner:

- Yellow warning: "X invoices are past 20 hours and close to the 24-hour deadline"
- Red danger: "X invoices have passed the 24-hour upload deadline"
- Only shown when warning/expired count > 0

Controls:

- "Upload All Pending (N)" button — processes entire pending queue
- "Refresh" button — manual refresh
- Auto-refreshes every 15 seconds

Queue Table:

Column	Details
Invoice Ref	Monospace font
Buyer	Business name
Queued	Time since queued (e.g. "2h 15m ago") + exact timestamp
Deadline	Badge + deadline timestamp + warning timestamp
Attempts	Retry count
Status	Badge
Actions	"Retry" button (only on failed/expired items)

Row highlighting:

- Red background: 24h deadline passed
- Yellow background: 20h warning
- Normal: within deadline

Deadline Badge states:

- "Within deadline" (light/neutral)
- "20h warning" (yellow)
- "24h deadline passed" (red)

Status Badge states:

- SUBMITTED (green)
- UPLOAD FAILED (red)
- OFFLINE / PENDING (gray)
- DEADLINE PASSED (dark)

UX notes:

- This page is urgency-critical — operators must upload within 24h by law
- Auto-refresh is implemented (15s interval)
- Retry is available on UPLOAD_FAILED and expired items
- The "Upload All Pending" button is disabled when queue is empty

3.9 Company Profile (/company-profile)

Purpose: Store the company's (seller's) details. These auto-fill the seller fields in every new invoice.

Header section:

- Company logo (circular, 80×80) — shows uploaded logo or default icon
- Company name + email (below logo)
- "Edit" button (view-only by default)

Edit mode:

- "Edit" button switches all fields to editable
- "Save" and "Cancel" buttons replace "Edit"

Form Fields:

Field	Type	Options
Company Name	Text	
NTN or CNIC	Text	
Business Type	Select	Retail, Wholesale, Manufacturing, Service
Province	Select	Punjab, Sindh, Khyber Pakhtunkhwa, Balochistan, Gilgit Baltistan
Address	Text	
Phone Number	Text	
Company Logo	File upload (image)	Only shown in edit mode

Email Address section:

- Displayed separately from the main fields
- Has its own "Edit Email Address" button
- Clicking opens a dedicated email edit modal (separate from main form)

Email Edit Modal:

- Single email input
- Save / Cancel

Notification:

- Auto-dismiss banner (3s) for success/error after save

UX notes:

- Fields are disabled/grayed in view mode
- Logo is uploaded as base64, stored server-side
- The company profile is the single source of truth for seller info — must be filled before invoice submission

3.10 Customers (/customers)

Purpose: Address book for buyer businesses. Used to quick-fill buyer fields in invoice form.

Header Stats Row (2 cards):

- "No of Customers" count (with people icon)
- "No of Screens" / "Digital" count (decorative stat)

Controls:

- "Add Customer" button → opens Add modal
- Search input (filters by name, CNIC, province, address, registration type)
- Date filter: From / To date range

Table Columns:

Column	Notes
S.No	Row number
Customer Name	
CNIC	
Phone	
Province	
Registration Type	
Actions	View, Edit, Delete icons

Modals (Bootstrap):

- **Add Customer Modal:** name, CNIC, phone, email, province, address, registration_type
- **Edit Customer Modal:** same fields pre-filled
- Delete: `window.confirm` dialog (no custom modal currently)

Pagination: 5 rows per page, numbered buttons

UX notes:

- Province options: Punjab, Sindh, Khyber Pakhtunkhwa, Balochistan, Gilgit Baltistan
- Registration Type: Registered / Unregistered (used in FBR invoice submission)
- Customers link to invoice buyer data for dashboard derivation

3.11 Products (/products)

Purpose: Product catalog with HS code mappings. Used to auto-fill invoice line items.

Controls:

- "Add Product" button → opens modal
- Search input (by name, HS code, description)
- Sort dropdown: Newest / Oldest / Name A-Z / Name Z-A / Price High-Low / Price Low-High / Stock High-Low / Stock Low-High

Table Columns:

Column	Notes
Tracking No	Product ID
Product Name	
Sales Tax Rate	Shown as percentage
In Stock	
HS Code	
UOM	
Status	Active (green) / Inactive (gray) badge
Action	View, Edit, Delete buttons

Product Modal (Add/Edit/View):

- Mode: "Add Product" / "Edit Product" / "View Product" (read-only)
- All fields editable in Add/Edit, disabled in View

Modal Form Fields:

Field	Type	Notes
Product Name	Text	
Sales Tax Rate (%)	Number	
Unit of Measure	Select	Loaded from FBR UOM reference
HS Code	Select	Loaded from FBR item descriptions (HS codes)
In Stock	Text	
Description	Textarea	Auto-filled when HS code is selected
Sale Type	Select	Loaded from FBR transaction types
SRO Schedule No	Text	Optional
Further Tax Applicable	Checkbox	
Extra Tax Applicable	Checkbox	
Status	Select	Active / Inactive

Pagination: 5 rows per page, numbered buttons

UX notes:

- FBR reference data (HS codes, UOM, transaction types) loaded from bootstrap API on mount
- Falls back to hardcoded options if API unavailable
- Selecting HS code auto-fills Description
- Modal uses Bootstrap JS (`window.bootstrap.Modal`)

3.12 Staff (/staff)

Purpose: Staff directory. (Internal record-keeping; not directly used in invoice submission.)

Table Columns: Staff Name, Designation, CNIC/NTN, Phone, Province, Actions (View/Edit/Delete)

Modal Fields: Member Name, Designation, CNIC/NTN, Phone Number, Email, Province, Address

Pagination: 5 rows per page

3.13 Services (/services)

Purpose: Informational/marketing page listing services offered. Static or lightly dynamic content.

3.14 Sandbox Testing (/sandbox)

Purpose: Run FBR test scenarios to certify the integration. Required before PRAL grants a production token.

Summary Cards Row (6 cards):

Card	Color
Total Scenarios	Neutral
Ready	Blue
Passed	Green
Failed	Red
Not Run	Yellow
Placeholder	Gray

PRAL Certification Progress Bar:

- Shows passed / ready as a percentage
- Bar is green if no failures, red if any failed
- Success message: "✓ All ready scenarios passed — eligible for PRAL production token request"

Controls:

- Operation selector: "Submit (recommended)" / "Validate only"
- "Run All Scenarios (N ready)" button
- "Refresh" button
- "Clear Results" button (danger, right-aligned)

Scenarios Table:

Column	Notes
--------	-------

Column	Notes
ID	Monospace (e.g. SN001)
Scenario	Name + description
Category	"general" (blue) / sector-specific (cyan)
Status	Passed / Failed / Not Run / Placeholder badge
FBR Invoice No	Result invoice number
Last Run	Relative time (e.g. "2h ago")
Actions	► Run button + Details toggle

Expandable Detail Row:

- Operation type, status code, duration (ms)
- Error list (monospace, red) if failed

Row highlighting: Red background for failed rows.

Status Badge states:

- Passed (green)
- Failed (red)
- Not Run (yellow)
- Placeholder (gray, disabled)

UX notes:

- 9 required scenarios must pass before production eligibility: SN001, SN002, SN005, SN006, SN007, SN008, SN016, SN026, SN027
- Placeholder scenarios are disabled (not yet implemented — awaiting FBR test data)
- Running all scenarios is a destructive-ish operation; "Clear Results" requires confirm dialog

3.15 FBR Settings (/settings)

Purpose: Configure the FBR API integration — environment, tokens, mock mode.

Layout: Two-column (form left, status cards right)

Left Column — Configuration Form:**Environment selector (toggle buttons):**

- Sandbox () / Production ()
- Description text changes based on selection

Mock Mode toggle (switch):

- ON → "Mock Mode ON" (yellow highlight box): no real FBR calls, fake invoice numbers generated locally
- OFF → "Live Mode ON" (green highlight box): real FBR API called, valid token required

Sandbox API Token:

- Password input (hidden by default)
- Shows masked current value if already configured (e.g. Bearer ****abc123)
- Placeholder: "Paste sandbox token here" or "Enter new token to replace"
- Optional: "Remove stored sandbox token" checkbox (danger label, shown if configured)

Production API Token:

- Same pattern as sandbox token

Action Buttons:

- "Save Settings" (primary)
- "Verify Token with FBR" (secondary) — pings FBR with current token

Right Column — Status Cards (3 cards):**Token Status Card:**

- Sandbox status badge + message
- Production status badge + message
- Active token badge + message + "Checked at" timestamp
- "Refresh" button

Status Badge values:

Status	Label	Color
mock	"Mock Mode"	Gray
missing	"Not Configured"	Red
configured_unverified	"Saved (unverified)"	Yellow
active	"Active"	Green
invalid	"Invalid Token"	Red
error	"Error"	Red

Outbound IP Address Card:

- Displays server's public IP in a dark code block
- "Copy" button → copies to clipboard, shows "✓" briefly
- Explanation: "Submit this IP to PRAL for whitelisting before going live"
- Shows source and local addresses

Current State Summary Card:

- Active environment badge (blue)
- FBR mode badge (Mock=yellow, Live=green)
- Sandbox token status (Configured=green, Missing=red)
- Production token status
- "Last saved" timestamp

Success/Error Alerts: Dismissible banners at top of page.

UX notes:

- Settings page is for operators/admins only — not end-users
- Token values are never shown in plaintext — stored encrypted on backend
- Mock mode is the default for development; switching to live requires a valid PRAL token

3.16 Support (/support)

Purpose: Help/support contact page. Static or form-based.

3.17 Account Sub-Pages (/account/*)

These are sub-pages accessed from the user's account area (likely accessible from a profile icon, not the main sidebar).

Sub-pages:

Path	Purpose
/account/edit-profile	Edit user's name, phone, profile info
/account/change-password	Change account password (current + new + confirm)
/account/two-factor	Toggle two-factor authentication
/account/language	Language preference (Google Translate integration)
/account/privacy	Privacy settings / policy
/account/terms	Terms of service
/account/deactivate	Deactivate account flow

These pages use their own AccSidebar component (separate from the main sidebar).

4. Global UX Patterns

4.1 Status Badges

Consistent badge system across screens:

Status	Color	Used on
SUBMITTED	Green	Invoices, Queue
FAILED / UPLOAD FAILED	Red	Invoices, Queue
DRAFT	Gray	Invoices
OFFLINE / PENDING	Gray	Queue
Active	Green	Products, Tokens
Inactive	Gray	Products
Passed	Green	Sandbox
Failed	Red	Sandbox
Not Run	Yellow	Sandbox
Mock Mode	Gray	Settings
20h Warning	Yellow	Queue deadline
24h Deadline Passed	Red	Queue deadline

4.2 Loading States

Every data fetch shows a spinner. Buttons show inline spinners while loading. Tables show a full-row loading message.

4.3 Empty States

Every table has a "No X found" message when data is empty.

4.4 Modals

Used for: Add/Edit/View records (Customers, Products, Staff), Invoice detail, Delete confirmation, Email edit.

Pattern: Bootstrap modal, header title changes (Add/Edit/View), footer has Cancel + Save/Update buttons, error displayed inline in modal body.

4.5 Toast Notifications

Success/failure toast messages (bottom-right) for all API operations. Library: react-toastify.

4.6 Pagination

Two patterns used:

- Numbered page buttons (Products, Customers, Staff — 5 rows/page)
- Prev/Next with page indicator (Invoice list — configurable rows/page)

4.7 Search / Filter

- Free-text search on: Invoices, Products, Customers
- Date range filter on Customers
- Sort dropdowns on Invoices and Products

4.8 Back Button Blocking

All authenticated pages block the browser back button (useBlockBackButton hook). Users must navigate via sidebar.

4.9 Online/Offline Detection

- useOnlineStatus hook monitors network
- Offline banner appears automatically
- Invoice submission switches to "Save Offline" mode when offline
- When connectivity returns, offline queue auto-processes

4.10 Language / Translation

Google Translate integration supports: English, Urdu, Spanish, French, Arabic. The translate widget is hidden; language is changed programmatically. A language preference is stored in localStorage.

5. Data & API Dependencies Per Screen

Screen	Key API Calls
Dashboard	/api/dashboard/invoices, /api/products, /api/dashboard/charts
Invoice List	/api/dashboard/invoices, /api/customers

Screen	Key API Calls
Add Invoice	/api/ref/bootstrap, /api/products, /api/customers, /api/company-profile, /api/sandbox/status, /api/invoice/submit, /api/queue/add
Upload Invoice	(file parsing, then same as Add Invoice)
Offline Queue	/api/queue, /api/queue/status, /api/queue/:id/retry
Company Profile	/api/company-profile (GET + PUT)
Customers	/api/customers (CRUD)
Products	/api/products (CRUD), /api/ref/bootstrap
Staff	/api/staff (CRUD)
Sandbox	/api/sandbox/status, /api/sandbox/run/:id
Settings	/api/settings, /api/token/status, /api/token

6. Key Design Considerations for Redesign

6.1 Most Critical Screens (prioritize these)

1. **Add Invoice** — the core action, most complex form, most frequently used
2. **Invoice List** — primary day-to-day view; needs filtering + download
3. **Offline Queue** — compliance-critical; needs clear urgency indicators
4. **Settings** — admin-critical; complex status display

6.2 Complexity Notes for Add Invoice

- The line items section needs to handle dynamic rows (add/remove)
- HS code is a large dropdown (thousands of codes) — consider a searchable select / autocomplete
- Many fields are interdependent: selecting product fills 6+ fields; selecting HS code fills description
- Mobile usability is a challenge for this form — consider multi-step wizard for mobile

6.3 Compliance Urgency (Offline Queue)

The 20h/24h deadline indicators are legally required to be visible. The UI must make these impossible to miss — color coding alone may not be sufficient; consider alert banners, bold countdown timers, or notification badges in the sidebar.

6.4 Sidebar Navigation

- Invoice has sub-items (Add, Upload, Offline Queue)
- Sub-items show/hide when Invoice is expanded
- Collapsed sidebar should still indicate active section

6.5 PDF Invoice

- Generated client-side (jsPDF)
- Includes: FBR DI logo, seller info, buyer info, line items table, QR code (FBR Invoice Number)
- This is a legal document — layout must be clean and formal
- Current size: A4 portrait

6.6 FBR Error Messages

The `FbrErrorDisplay` component shows structured errors from FBR API. These should be clearly distinguishable from form validation errors (field-level vs. submission-level).

6.7 Current Known UI Gaps (not yet implemented)

5. **Dashboard chart** — states and API wired, JSX rendering incomplete
6. **Invoice list date range filter** — search exists but no date filter yet
7. **Invoice list invoice type filter** — no filter by type yet
8. **Social login** (Google/Apple/Microsoft) — buttons present, no functionality
9. **Sign up flow** — link present, no flow
10. **Support page** — placeholder
11. **Services page** — placeholder/marketing content

7. Image & Asset Inventory

Asset	Usage
Images/New Black Logo 1.png	Main app logo (sidebar, login)
Images/TechionikIcon.png	Small icon (sidebar collapsed, footer)
Images/imglogin.png	Login page illustration (right panel)
Images/fbr-di-logo-crop.png	FBR Digital Invoicing logo (used in invoice PDF)
Images/google.png	Google login button icon
Images/apple.png	Apple login button icon
Images/microsoft.png	Microsoft login button icon

8. Glossary of Domain Terms

Term	Meaning
FBR	Federal Board of Revenue (Pakistan tax authority)
PRAL	Pakistan Revenue Automation Ltd (operates FBR gateway)
NTN	National Tax Number (business tax ID)
CNIC	Computerized National Identity Card number (individual ID)
HS Code	Harmonized System code (commodity/product classification)
UOM	Unit of Measurement
SRO	Statutory Regulatory Order (tax rate schedule)
Sale Invoice	Standard taxable invoice
Debit Note	Invoice for additional charges

Term	Meaning
Credit Note	Invoice for refunds/reductions
Sandbox	FBR test environment (no real tax records created)
Mock Mode	Local simulation (no FBR contact at all)
Rule 150R	FBR rule requiring 26 mandatory invoice fields
Rule 150XC	FBR rule requiring offline invoices uploaded within 24 hours
Scenario ID	FBR-assigned test case identifier (e.g. SN001)
eligibleForProduction	Flag indicating all 9 required sandbox scenarios have passed

End of Frontend Requirements Report